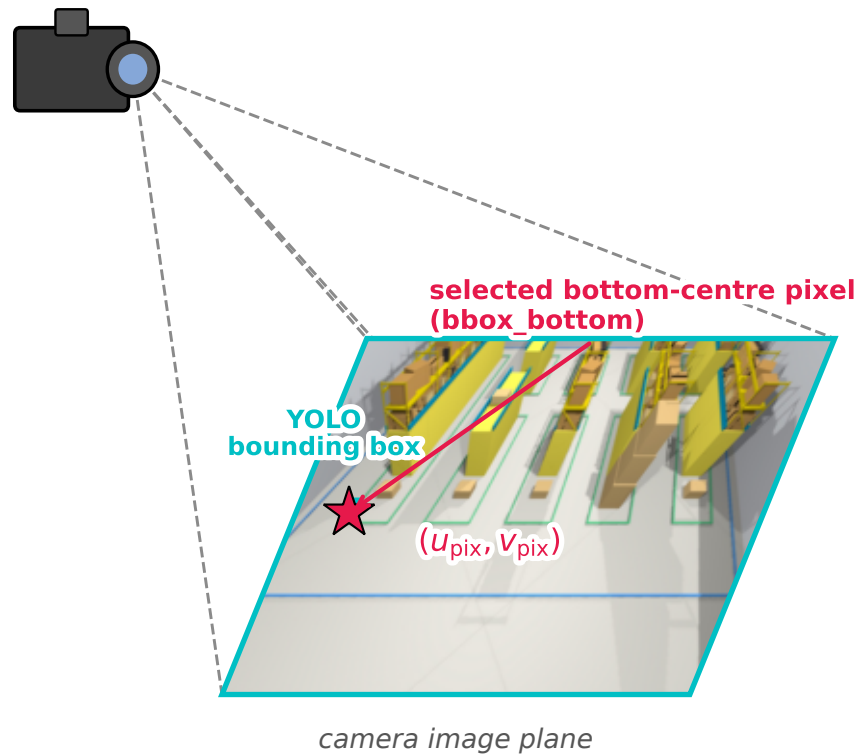


fixed camera

(a) YOLO image observation



(b) projection model

ground-plane
homography



$$[\tilde{x}, \tilde{y}, \tilde{w}]^T = H^{-1}[u_{\text{pix}}, v_{\text{pix}}, 1]^T$$

$$(\hat{x}, \hat{y}) = (\tilde{x}/\tilde{w}, \tilde{y}/\tilde{w})$$

(c) ground-plane estimate

back-projected point: $(\hat{x}, \hat{y})$

